# StockSense AI

Sales forecasts and stock suggestions for a fictional mini market

Store 1  5 products  7-day forecast

# The weekly stock decision

## How much should the manager order before next week’s sales are known?

### Too little stock

Missed sales

### Too much stock

Money tied up in inventory

# A small, clear prototype

**1**

fictional store

**5**

anonymous products

**7**

forecast days

The app turns these forecasts and entered stock values into an order suggestion.

# The sales data

**913,000**

rows in the source file

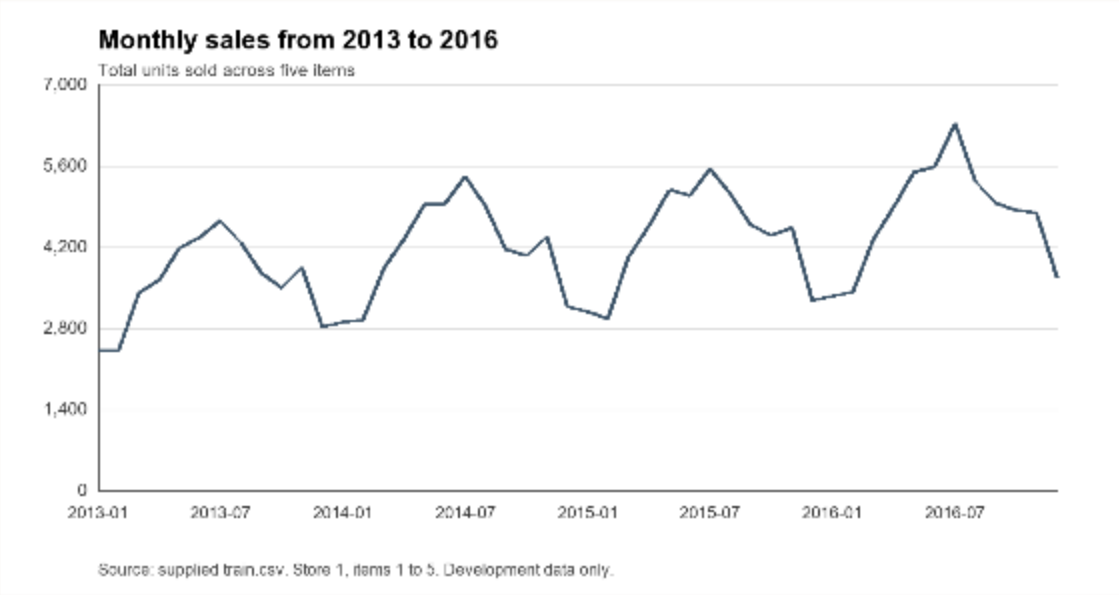
**9,130**

rows in our selected store and items

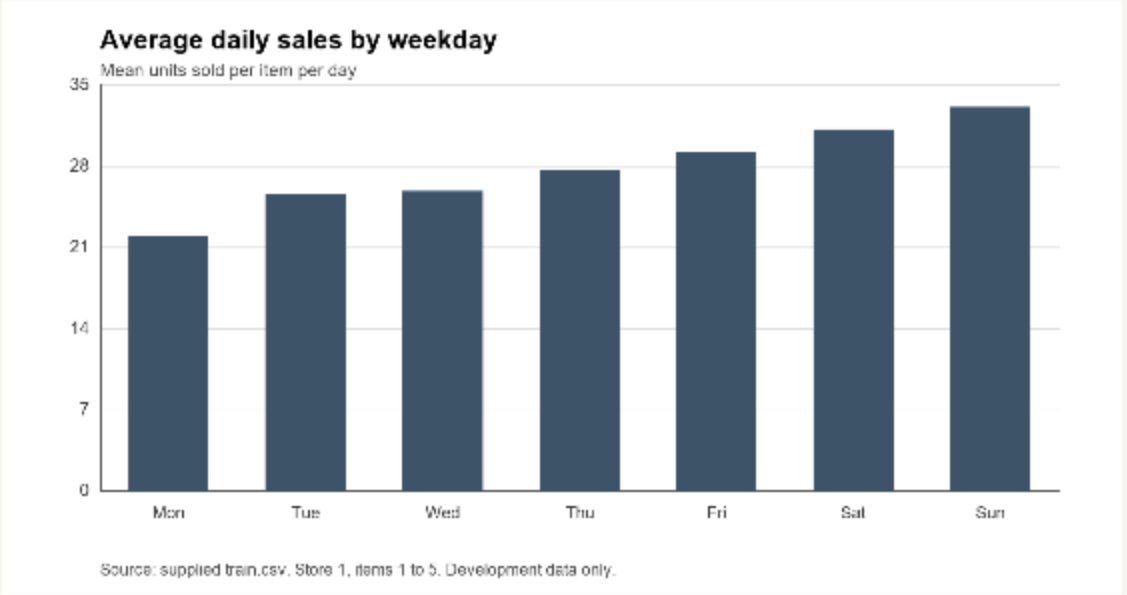
## Daily sales from 2013 to 2017

No product names, stock levels, prices or supplier records are supplied.

# Sales patterns before testing




## Monthly sales totals




## Average by weekday

# Time split and methods

## 2013–15

Fit the models

## 2016

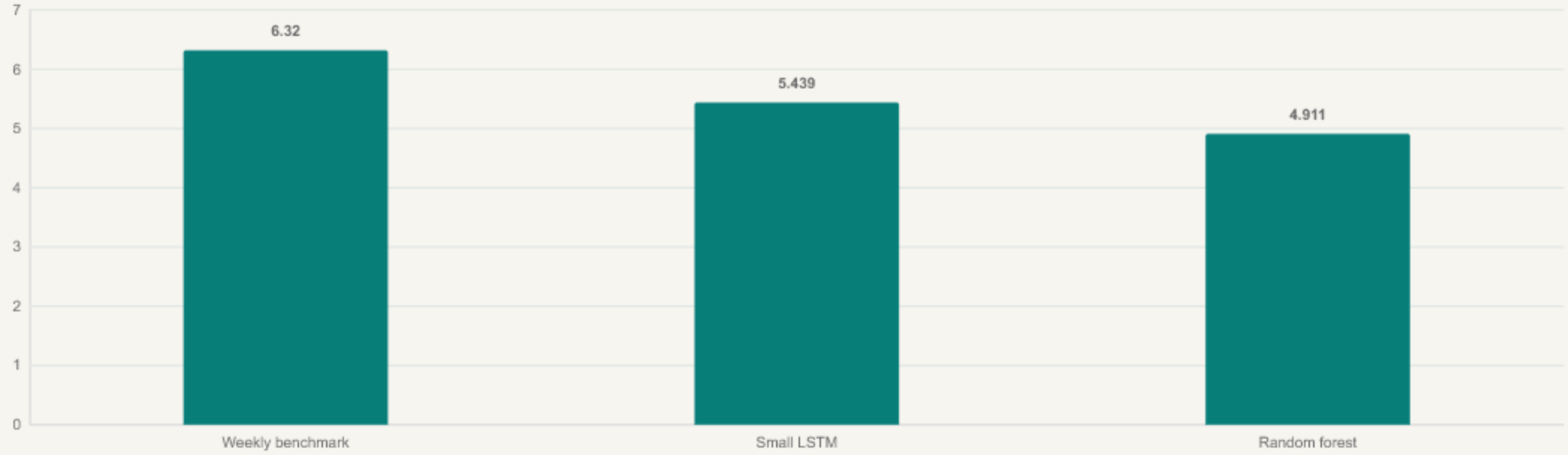
Select by daily MAE

## 2017

Final test

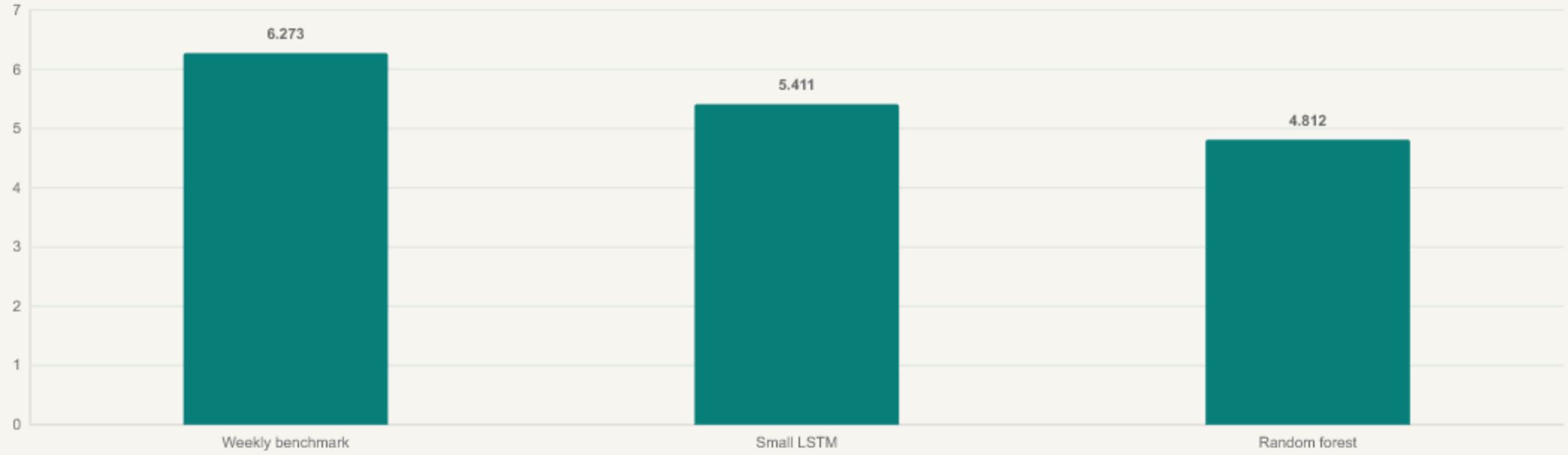
Compared: last-week repeat, random forest and a small LSTM

# Random forest won validation




Daily mean absolute error (units)

# The result held on the final test




## 23.3% lower daily error than the weekly benchmark

LSTM had the lowest weekly-total error (17.343 units).

# A small LSTM comparison

**28**

past sales days

**16**

LSTM units

**7**

predicted days

It had better weekly totals overall, while random forest had lower daily error.

# How the order number is calculated

**77.52**

forecast units during coverage

**+ 10 buffer = 87.52 target**

50 available, 0 outstanding

**Suggest ordering 38 units**

# Ask StockSense

## “What will Product 1 sell next week?”

The assistant identifies the request type, then reads the saved forecast.

### 5 supported request types

Forecast, order, recent sales, model results and help

# Prototype checks

**52**

automated tests passed

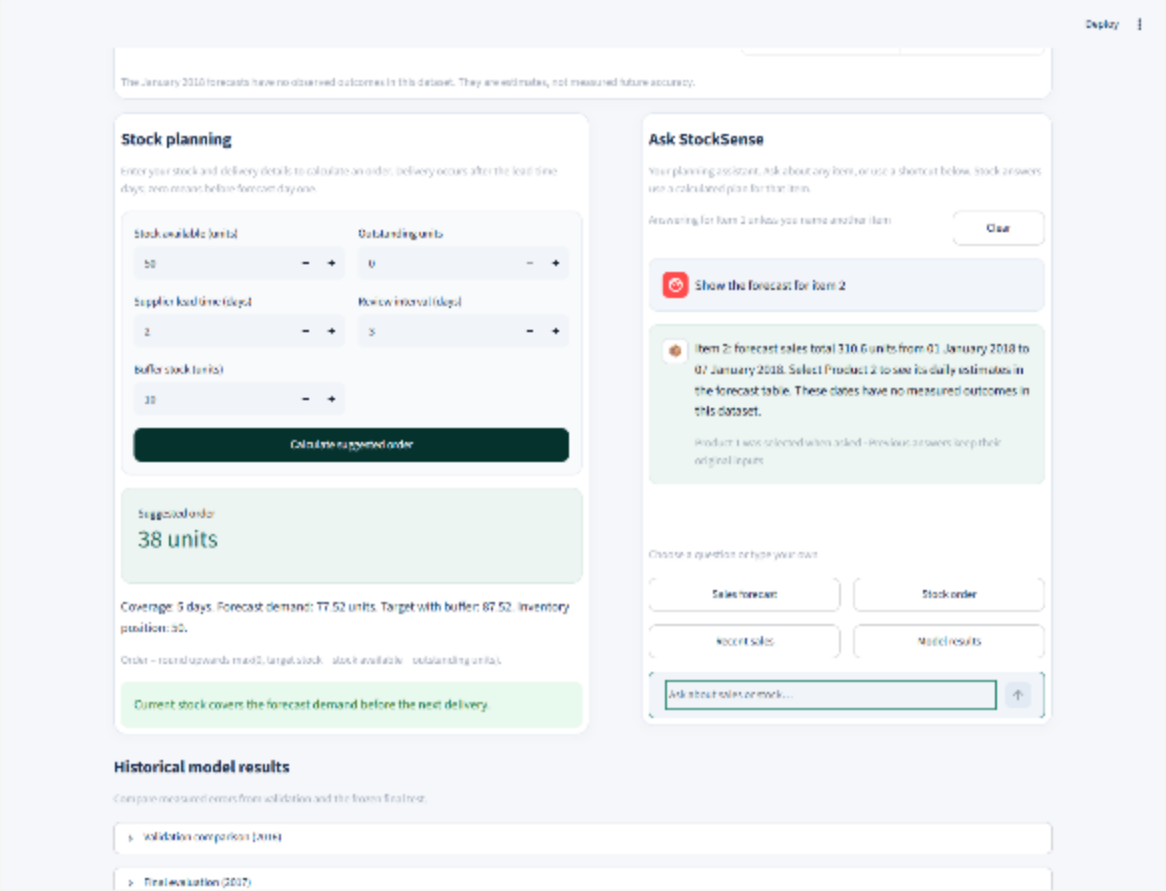
**24/25**

supported requests recognised

**8/8**

unrelated requests rejected

# Live prototype walkthrough




01 Select a product

02 View seven days

03 Enter stock

04 Ask a question

# What we can conclude

## Random forest had the lowest daily test error.

The app turns forecasts into a clear order calculation.

Real savings and January 2018 accuracy remain unmeasured.

## Questions